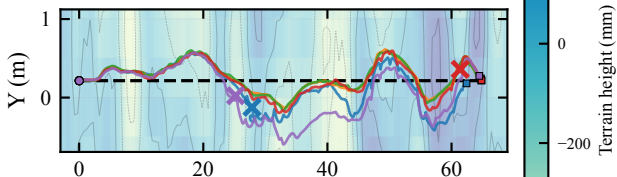
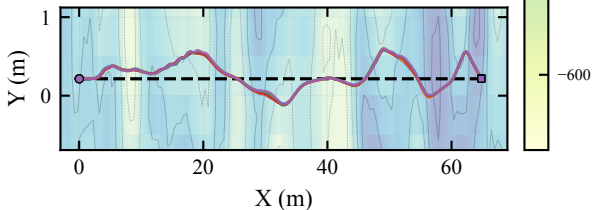


(a) Without attitude control ($\bar{e} = 36.48$ m)



(b) Cascaded PID ($\bar{e} = 0.48$ m)



(c) PID + residual RL ($\bar{e} = 0.26$ m)